

Immunosuppressive Conditions and Risk Factors for Herpes Zoster

Immunosuppressive conditions associated with high risk of herpes zoster include human immunodeficiency virus (HIV) infection, bone marrow transplant, leukemia and lymphoma, use of cancer chemotherapy, and use of corticosteroids. Herpes zoster is a prominent and early "opportunistic infection" in persons infected with HIV, in whom it is often the first sign of immune deficiency. Thus, HIV infection should be considered in individuals who develop herpes zoster.

Other factors reported to increase the risk of herpes zoster include female sex, physical trauma in the affected dermatome, interleukin-10 gene polymorphisms, and black race, but confirmation is required. Exposure to children and contact with cases of varicella have been reported to confer protection against herpes zoster.

Second episodes of herpes zoster are uncommon in immunocompetent persons, and third attacks are very rare. Persons suffering more than one episode may be immunocompromised. Immunocompetent patients suffering multiple episodes of herpes zoster-like disease are likely to be suffering from recurrent zosteriform herpes simplex virus (HSV) infections.

Contagiousness and Transmission

Patients with herpes zoster are less contagious than patients with varicella. The rate at which susceptible household contacts develop varicella after exposure to herpes zoster appears to be about one-third of the rate observed after exposure to varicella. Virus can be isolated from vesicles and pustules in uncomplicated

herpes zoster for up to 7 days after the appearance of the rash, and for much longer periods in immunocompromised individuals.

Patients with uncomplicated dermatomal zoster appear to spread the infection by means of direct contact with their lesions. Patients with disseminated herpes zoster may, in addition, transmit the infection via aerosols, so that airborne precautions, as well as contact precautions, are required for such patients.

Impact of Varicella Vaccination on Herpes Zoster Epidemiology

The effect of the marked reduction in the incidence of varicella, due to widespread varicella vaccination of children, on the epidemiology of herpes zoster is unclear. In the long term, the incidence of herpes zoster is likely to decline as the cohorts of children now receiving the vaccine become adults; vaccine virus-associated herpes zoster will probably be less frequent and less severe in older adults than wild-type virus-associated herpes zoster because the vaccine virus is highly attenuated.

In the short term, the incidence of herpes zoster could increase because a decline in the incidence of varicella will reduce the adult population's exposure to VZV, thereby reducing immune boosting, hastening the age-related decline in immunity to VZV, and thus increasing the age-specific risk of herpes zoster. However, recent studies of herpes zoster in populations with high rates of varicella vaccination have shown little or no increase in the incidence of herpes zoster.

PATHOGENESIS

Varicella-zoster virus (VZV) is a member of the herpesvirus family. Other members pathogenic for humans include herpes simplex virus type 1 (HSV-1) and type 2 (HSV-2), cytomegalovirus, Epstein-Barr virus, human herpesvirus-6 (HHV-6) and HHV-7 (which cause roseola), and Kaposi sarcoma-associated herpesvirus (also called HHV-8). All herpesviruses are morphologically indistinguishable and share key properties, including the ability to establish latent infections that persist for life.

The VZV genome encodes approximately 70 unique genes, most of which have DNA sequence and functional homology to genes of other herpesviruses. Immediate-early gene products regulate VZV replication. Early gene products, such as the virus-specific thymidine kinase and the viral DNA polymerase, support viral replication. Late genes encode structural proteins of the virus that serve as targets for antibody and cellular immune responses.

There is only one VZV serotype. Although viruses isolated from individual cases of varicella or herpes zoster worldwide are fundamentally similar, minor variations in their nucleotide sequences allow differentiation between wild-type and vaccine virus strains and enable "fingerprinting" of viruses from individual patients.

Pathogenesis of Varicella

VZV enters through the mucosa of the upper respiratory tract and oropharynx. Initial viral replication at this site leads to primary viremia, with dissemination of small amounts of virus via the blood and lymphatics. This virus is then cleared by cells of the reticuloendothelial system, which become the major site of viral replication during the remainder of the incubation period.